

Resolution-Enhanced Lensless Ptychographic Microscope Based on Maximum-Likelihood High-Dynamic-Range Image Fusion

Li Liu¹, Wenjie Li¹, Ming Gong¹, Lei Zhong¹, Honggang Gu¹, and Shiyuan Liu¹

Abstract—The Abbe diffraction limit tells that the resolution of an imaging system is completely determined by the numerical aperture (NA) for a coherent illumination wavelength. Here, a simple-to-implement and noise-robust maximum-likelihood high-dynamic-range (ML-HDR) image fusion approach is proposed to extend the effective NA of the lensless imaging microscope to enhance the resolution of the ptychography. The proposed approach constructs a joint probability to optimize the weighting function through the maximum-likelihood estimation, allowing the direct fusion of the HDR absolute irradiance from a sequence of diffraction frames, dark frames, and exposure times, and it needs no additional and complex parameter calibration and prior modulation templates. The ML-HDR image fusion approach considers mixed noises for the first time, making it proficiently retrieves high-frequency ptychographic signals without losing or reducing the diffraction signal-to-noise ratio (SNR) in low-frequency regions. A series of simulations and experiments both in transmission and reflection geometries are conducted compared with the single-exposure ptychography, and results indicate that the ML-HDR approach significantly mitigates 8 bits for the detector's dynamic range and enhances the resolution of ptychographic imaging nearly threefold. Compared with existing HDR techniques based on the linear response function calibration, the proposed ML-HDR demonstrates superior noise immunity and higher reconstruction qualities. These remarkable advancements may greatly broaden

the applications of the ML-HDR image fusion approach in lensless computational imaging fields, including but not limited to ankylography, holography, and tomography.

Index Terms—Computational imaging, high dynamic range (HDR), image fusion, maximum likelihood, ptychography, resolution enhancement.

I. INTRODUCTION

BENEFITING from inherent properties of non-interference and lensless advantages, the coherent diffraction imaging (CDI) has been applied to emerging nanomaterials [1], semiconductor physics [2], and biological cytology [3]. Meanwhile, emerging interdisciplinary fields put forward new demands for higher resolution and throughput imaging techniques. The Abbe diffraction limit tells that the resolution depends only on the numerical aperture (NA) and illumination wavelength of the imaging system [4]. Thus, in lensless CDI, where the wavelength remains invariant, the imaging resolution is entirely determined by the effective NA, i.e., the angular range from the weak high-frequency scattering not swamped by noise to the strong low-frequency scattering in Fourier space. Normally, the scattering intensity in the low-Q region of the near-zero angle is several orders of magnitude greater than that in the high-Q region in the reciprocal space. However, for pixel charges with the full well effect, the detector only records coherent diffraction fields within a certain dynamic range [5]. This limitation severely affects the signal-to-noise ratio (SNR) of diffraction fields with different photon fluxes and further impacts the effective NA and resolution of the diffraction imaging.

Resolution-enhancement techniques that broaden the dynamic range of the detector to improve the effective NA have been widely researched by scholars and mainly include two different research lines in lensless diffraction imaging. The first line of thought attempts to reduce the magnitude ratio of the diffraction intensities between the low-Q and high-Q regions in finite dynamic ranges by optimizing experimental structures, including beamstop methods [6], [11] and evacuating flight tube methods [7], [8], [9], [10], [11], [12]. Reinhardt et al. [6] compared scenarios with and without a beamstop in the central plane of detectors, aiming to reduce the non-ideal background scattering and to better collect low-dynamic-range diffraction patterns. The results showed a remarkable improvement in ptychographic sensitivity and spatial resolution. Meanwhile, Eschen et al. [7]

Manuscript received 2 September 2023; revised 12 December 2023; accepted 2 January 2024. Date of publication 8 February 2024; date of current version 21 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 52130504, in part by the Key Research and Development Program of Hubei Province under Grant 2021BAA013, in part by the Fundamental Research Funds for the Central Universities under Grant 2021XXJS113, in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2023A1515030149, and in part by the Innovation Project of Optics Valley Laboratory under Grant OVL2023PY003. The Associate Editor coordinating the review process was Dr. Yuya Koyama. (Corresponding authors: Honggang Gu; Shiyuan Liu.)

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Digital Object Identifier 10.1109/TIM.2024.3363788

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and Liu et al. [8] designed a special mask to modulate the orbital angular momentum entrance-wave for the extreme ultraviolet ptychography, which significantly improved ptychographic imaging quality. By combining the cylindrical reference object with the beamstop, Suzuki and Takahashi [9] proposed a “dark-field” X-ray ptychography, which reduced the dynamic range of the diffraction field by three orders of magnitude. Recently, a low-cost coherent modulation imaging (CMI) [10] has been proposed, which does not require any special designed patterns or strict installation position restrictions to modulate exit-wavefront. By combining the beamstop with the CMI, Liu et al. [11] extrapolated the lost data caused by the beamstop based on the modified modulus constraint, which further reduced the demand for the dynamic range of the detector while increasing the sparse data compared with conventional beamstop methods. Furthermore, Wang et al. [12] proposed a novel algorithm for the CMI utilizing unknown modulators, and made the CMI technologies becoming more versatile and robust for enhancing the resolution. In reviewing these modulation methods, whether beamstop, entrance-wave, reference-object, or exit-wavefront, although the introduction of modulators mitigates the demand for the dynamic range of detectors, it increases the complexity of the imaging system, which inevitably bring additional uncertainties or data loss.

In order to counteract the additional uncertainty and instability introduced by the above modulation methods, the other alternative line of thought emphasizes the acquisition of high-dynamic-range (HDR) diffraction fields that go beyond the original finite dynamic range of the detectors. The HDR approach works by multi-exposure fusion (MEF), which involves modifications of exposure times [13], [14] and color filters [15], and has been applied in fluorescence imaging [16], infrared imaging [17], and computational photography [18], [19], [20], [21], [22], [23], [24], [25]. For instance, Xu et al. [18] introduced an unsupervised and unified end-to-end MEF method, U2Fusion, for improving feature extraction and information measurement. In addition, Liu et al. [19] developed a transformer-based model equipped with a pyramid cross-attention alignment module that was specifically designed to aggregate highly correlated features derived from different exposure frames. These state-of-the-art MEF methods based on deep learning aim to extract more details [20] for visual display or measurement [21] and to search for the information entropy of the HDR imaging by tone mapping [22] and gradient domain operation [23]. It is hazardous for computational imaging, including ptychography, tomography, and holography, that seeks realistic HDR diffraction signals rather than information richness. Besides the deep learning methods, Oh et al. [24] and Li et al. [25] also respectively proposed the rank minimization and the structural patch decomposition methods based on linear response assumption to reconstruct the real HDR scenario, both of which have the different rejection pixel strategy for different fusion patterns. This non-unification optimization strategy breaks the overlap constraint for the ptychographic phase retrieval, which makes it unacceptable for the ptychographic raw signals fusion. Therefore, researchers have

to focus on the classical HDR image fusion method in lensless computational imaging. By calibrating the linear response function of the photoelectric sensor, Leong-Hoi et al. [26] and Liu et al. [27] first applied it to the tomography and the ptychography, respectively. Meanwhile, Tanksalvala et al. [28] applied this technique to the EUV phase-sensitive reflectometry for non-destructive characterization of three-dimensional nanostructures with chemical specificity. These classical HDR image fusion methods based on the linear-response-function calibration (LRFC) are inevitably coupled with detector noises. Each type of noise affects the final fusion accuracy, and the rejection pixel strategy also results in the loss of diffraction information. This highlights the ongoing challenge in enhancing the imaging resolution while maintaining the SNR of the collected raw correct data.

To reduce the effect of noises on diffraction signals before the ptychographic phase retrieval and reduce dependence on hardware for lensless computational imaging systems [5]. In this article, a simple-to-implement and noise-robust maximum-likelihood HDR (ML-HDR) image fusion method is presented to enhance the resolution in ptychographic microscope. In this approach, the weighting function is optimized by a maximum likelihood estimation to retrieve the HDR absolute irradiance from a series of direct measurement parameters, including diffraction frames, dark frames, and exposure times. Different from the beamstop and evacuating flight tube modulation techniques in diffraction imaging, the proposed multi-exposure ML-HDR approach does not require prior modulation devices or templates, which avoids increasing system uncertainties, and there is no data loss resulting from the beamstop. More importantly, for the first time, the ML-HDR method considers the influence of temporal dynamic mixed noises and without any pixel rejection operation to miss diffraction signals on the HDR image fusion process, which makes it more noise-immunity and simple-to-implement than the state-of-the-art LRFC-HDR method in ptychographic microscope. These advantages make the ML-HDR an ideal and general candidate for a wide range of lensless CDI applications with lower SNR scenarios such as ankylography, holography, and tomography.

II. METHODOLOGY

A. ML-HDR and LRFC-HDR Image Fusion

Whether it is a complementary metal-oxide-semiconductor (CMOS), charge-coupled device (CCD) or other detectors, the optical intensities are inevitably contaminated by noises during the acquisition process. Therefore, the establishment of a complete detector noise model is of great significance for the error tracing and the signal processing. During the exposure time t_i ($i = 1, 2, \dots, N$), the relationship between the digital readout Z_i and the optical irradiance of the detector can be expressed as [29]

$$Z_i = \begin{cases} 0 & 0 \leq E < E_{\text{threshold-min}} \\ g(Et_i + Dt_i + R) & E_{\text{threshold-min}} \leq E < E_{\text{threshold-max}} \\ Z_{\text{max}} & E_{\text{threshold-max}} \leq E \end{cases} \quad (1)$$

where E is the optical irradiance coupled with the photon shot noise (PSN), D is the dark current shot noise (DCSN) from photons to charges, R is the reading noise (RN) from charges to the digital readout, and g is the overall gain coefficient during the conversion process. Due to the inherent properties of the quantum efficiency and full well effect [30] of detectors, the digital readout can be recorded in a finite dynamic range between the minimum and maximum optical irradiances $[E_{\text{threshold-min}}, E_{\text{threshold-max}}]$. Under the influence of Poisson noises (i.e., the PSN and DCSN) and Gaussian noises (i.e., the RN), mathematical expectation μ_{Z_i} and variance $\sigma_{Z_i}^2$ of the digital readout can be calculated respectively, as given by

$$\mu_{Z_i} = g(\mu_E t_i + \mu_D t_i + \mu_R) \quad (2)$$

$$\sigma_{Z_i}^2 = g t_i (\mu_E + \mu_D) + g^2 \sigma_R^2 \quad (3)$$

where μ_E , μ_D , and μ_R are the mathematical expectations for the PSN, DCSN, and RN respectively, and σ_R^2 is the variance of the RN. Similarly, the digital readout B_i , the mathematical expectation μ_{B_i} and the variance $\sigma_{B_i}^2$ of the dark frames during the exposure time t_i without the optical irradiance can be calculated respectively by

$$B_i = g(D t_i + R) \quad (4)$$

$$\mu_{B_i} = g(\mu_D t_i + \mu_R) \quad (5)$$

$$\sigma_{B_i}^2 = g t_i \mu_D + g^2 \sigma_R^2. \quad (6)$$

A joint distribution between the diffraction frames, the dark frames, and the optical irradiance E can be established as given by

$$\begin{aligned} Y_i &= Z_i - B_i - g E t_i \\ &= g[t_i(E_{Z_i} - E) + t_i(D_{Z_i} - D_{B_i}) + (R_{Z_i} - R_{B_i})] \end{aligned} \quad (7)$$

where $(E_{Z_i} - E)$ and $(D_{Z_i} - D_{B_i})$ obey the Skellam distribution, and $(R_{Z_i} - R_{B_i})$ obeys the Gaussian distribution. When the optical irradiance, namely Poisson eigenvalue, is very large, the Skellam distribution can be approximated as the Gaussian distribution [31], [32]. Therefore, in this case, the mathematical expectation and variance of the joint distribution can be calculated as

$$\begin{aligned} \mu_{Y_i} &= 0 \\ \sigma_{Y_i}^2 &= 2g(\mu_E t_i + \mu_D t_i) + 2g^2 \sigma_R^2. \end{aligned} \quad (8)$$

In the above mathematical derivation of the joint distribution, the conditional probability is equal to the joint probability of the joint distribution, as given by

$$P(y_i | \mu_Y, \sigma_{Y_1}^2, \sigma_{Y_2}^2, \dots, \sigma_{Y_i}^2) = \prod_{i=1}^N P(y_i | \mu_Y, \sigma_{Y_i}^2) \quad (9)$$

and the maximum likelihood function of the absolute irradiance $\hat{\mu}_E$ can be expressed as

$$\begin{aligned} \hat{\mu}_E &= \arg \max_{\mu_E} \prod_{i=1}^N P(y_i | \mu_Y, \sigma_{Y_i}^2) \\ &\propto \arg \max_{\mu_E} \left(\exp \left(\sum_{i=1}^N \frac{(Z_i - B_i - g t_i E)^2}{-2\sigma_{Y_i}^2} \right) \right) \end{aligned} \quad (10)$$

where N is the number of exposures. According to the first-order partial derivative of log-likelihood functions ($\partial P / \partial E = 0$), the maximum likelihood estimation of the absolute irradiance can be derived as

$$\begin{aligned} \hat{\mu}_E &= \frac{\sum_{i=1}^N (t_i^2 / \sigma_{Y_i}^2) (Z_i - B_i) / (g t_i)}{\sum_{i=1}^N (t_i^2 / \sigma_{Y_i}^2)} \\ &= \frac{\sum_{i=1}^N w_i (Z_i - B_i) / (g t_i)}{\sum_{i=1}^N w_i}. \end{aligned} \quad (11)$$

It's obvious that the (11) is a weighted-normalization expression, where the weighting factor w_i is

$$w_i = \frac{t_i^2}{\sigma_{Y_i}^2} = \frac{t_i^2}{g t_i \mu_E + \sigma_{B_i}^2}. \quad (12)$$

On the other hand, from the diffraction frames and dark frames of (1) and (4), we can approximately estimate the mathematical expectations of the absolute irradiance, as given by

$$\mu_E = \sum_{i=1}^N \frac{Z_i - B_i}{N g t_i}. \quad (13)$$

When further taking the response characteristics of the detector at different exposure times into account of the (6) and (13), and the weighting factor w_i of the (12) can be optimized as

$$w_i = t_i^2 / \left(t_i \sum_{i=1}^N \frac{Z_i - B_i}{N t_i} + \sigma_{B_i}^2 \right). \quad (14)$$

Through further in-depth analysis of the weighting factor w_i in the (14), we find that the weighting function suppresses the signal proportion under low exposure time. Therefore, for the case of low exposure time, although the Skellam distribution cannot be approximated simply as the Gaussian distribution, the optimized weighting function remains the physical applicability. Furthermore, when combining with different gain coefficients g , the HDR absolute irradiance of coherent different signals can be retrieved, as described as

$$E_{\text{HDR}}^{\text{ML}} = g \hat{\mu}_E = \frac{\sum_{i=1}^N w_i (Z_i - B_i) / t_i}{\sum_{i=1}^N w_i}. \quad (15)$$

Correspondingly in the LRFC-HDR image fusion, by alignment operation and ignoring the crosstalk of noises, the detector response function can be calibrated for different exposure-time diffraction frames as [26]

$$k_i = a i + b \quad (16)$$

where i is the exposure time, and a and b are the gain coefficient and bias coefficient of the linear response function, respectively. Rejecting overexposed pixels with different exposure times based on the camera's response function, the HDR absolute irradiance of coherent different signals with enhanced dynamic range can be fused as [27]

$$E_{\text{HDR}}^{\text{LRFC}} = \sum_i (k_i - B_i). \quad (17)$$

Here, $\sum_i$ represents the fusion process of overexposed pixels at i exposure time with underexposed pixels below i exposure time based on the linear response function. And B_i is the dark frames under i exposure time.

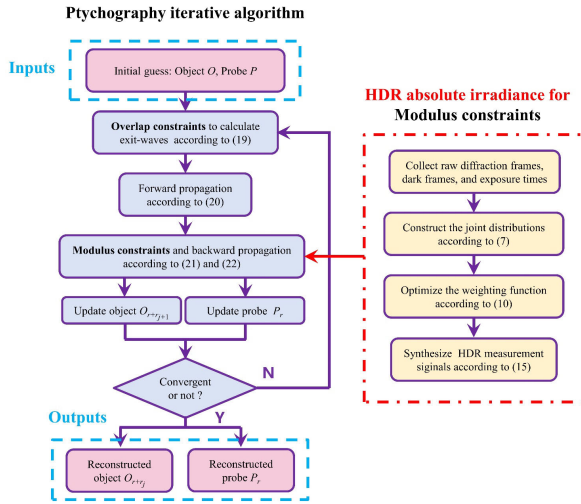


Fig. 1. Iterative flowchart of the ML-HDR ptychography.

B. Ptychographic CDI

After the HDR absolute irradiance E_{HDR} obtaining, the ptychographic non-convex optimization problem [33] can be finally optimized as

$$\Omega = \arg \min_{P_r, O_{r+r_j}} \sum_j \|E_{\text{HDR}} - F[P_r \times O_{r+r_j}]\|^2 \quad \forall r_j \in S \quad (18)$$

where P_r and O_{r+r_j} denote “probe” and “object” respectively, r is the real-space coordinate, r_j is the j th relative scanning offset (x_j, y_j) in the finite set S , and F indicates the 2-D scalar diffraction model. The more details of ptychographic phase retrieval reconstruction algorithm can be described as following in Fig. 1.

- 1) Raw coherent diffraction signals measurement. Raw coherent diffraction signals E_{HDR} at all scanning positions of the finite set S are measured by CCD or CMOS detectors.
- 2) Initial randomization guess of the probe P and the object O to be measured.
- 3) Make the finite set S scanning offsets. Probe-object interaction at different scanning positions generate exit-wave ψ_r as

$$\psi_r = P_r \times O_{r+r_j} \quad \forall r_j \in S. \quad (19)$$

Diffraction signals between different scanning positions form overlapping constraints in the real-space, and a large amount of redundant data between overlapping scanning positions effectively removes the ambiguous solution of ptychographic phase retrieval.

- 4) The exit-waves propagate through 2-D scalar diffraction model F , and the estimated coherent diffraction signals are obtained as

$$\Phi_q = F[\psi_r] \quad \forall r_j \in S \quad (20)$$

where q is the reciprocal-space coordinate. The scalar diffraction models include angular spectrum propagation, Fresnel diffraction, and Fraunhofer diffraction.

Algorithm 1 ML-HDR Ptychography

Input: Z_i, B_i, t_i and N ;

$$w_i = t_i^2 / \left(t_i \sum_{i=1}^N (Z_i - B_i) / (N t_i) + \sigma_{B_i}^2 \right);$$

$$E_{\text{HDR}}^{ML} = g \hat{\mu} E = \left(\sum_{i=1}^N w_i (Z_i - B_i) / t_i \right) / \left(\sum_{i=1}^N w_i \right); \quad \% \text{ HDR signal fusion}$$

Random initial: O and P

$M = \text{rand}(S)$ % Overlapping constraints

for $j = 1 : M$;

$$\psi_r = P_r \times O_{r+r_j};$$

$$\Phi_q = F[\psi_r];$$

$$\Phi'_q = \sqrt{E_{\text{HDR}}^{ML} \times \Phi_q / |\Phi_q|}; \quad \% \text{ modulus constraints}$$

$$\psi'_r = F^{-1}[\Phi'_q];$$

$$^{j+1} O_{r+r_j} = ^j O_{r+r_j} + ^j P_r^* / \left(\alpha |^j P_r|_{\max}^2 + (1-\alpha) |^j P_r|^2 \right) \times (\psi'_r - \psi_r);$$

$$^{j+1} P_r = ^j P_r + ^j O_{r+r_j}^* / \left(\beta |^j O_{r+r_j}|_{\max}^2 + (1-\beta) |^j O_{r+r_j}|^2 \right) \times (\psi'_r - \psi_r);$$

end

Output: O_{r+r_j} and P_r

- 5) The modulus constraint in reciprocal-space. Replace the estimated coherent diffraction signals with the measured HDR absolute irradiance and keep the phases unchanged.
- 6) The updated coherent diffraction signals Φ'_q can be obtained as

$$\Phi'_q = \sqrt{E_{\text{HDR}}} \times \frac{\Phi_q}{|\Phi_q|}. \quad (21)$$

- 7) The updated coherent diffraction signals are propagated by the inverse scalar diffraction diffraction F^{-1} , the updated exit-waves ψ'_r are calculated as

$$\psi'_r = F^{-1}[\Phi'_q] \quad (22)$$

the object to be measured and the probe based on exit-waves before and after the update at the j th position in the set S as

$$\begin{aligned} ^{j+1} O_{r+r_j} &= ^j O_{r+r_j} + \frac{^j P_r^*}{\alpha |^j P_r|_{\max}^2 + (1-\alpha) |^j P_r|^2} (\psi'_r - \psi_r), \\ ^{j+1} P_r &= ^j P_r + \frac{^j O_{r+r_j}^*}{\beta |^j O_{r+r_j}|_{\max}^2 + (1-\beta) |^j O_{r+r_j}|^2} (\psi'_r - \psi_r) \end{aligned} \quad (23)$$

where α and β are the step sizes of the ptychography. In general, β is larger than α [34] in the iteration process.

- 8) Take the updated object and probe as the initial guess of the $j+1$ th position, and repeat steps (3)–(6) until all scanning positions in the set S have been traversed, in which case only one iteration is completed.
- 9) Repeat iterations M times until the algorithm converges. The complete algorithm of the ML-HDR ptychography can be summarized as **Algorithm 1**.

Under the dual effect of modulus constraints and overlapping constraints [35], [36], a series of diffraction signals containing redundant data are frequently iterated to retrieve phase until the amplitude and phase information of the object to be measured are reconstructed in ptychographic

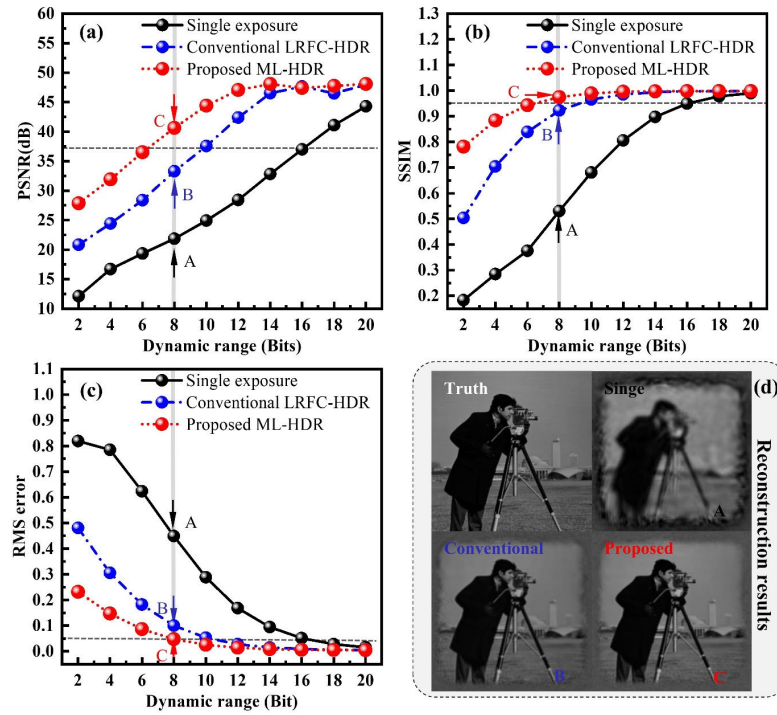


Fig. 2. Comparison of the single exposure, the conventional LRFC-HDR and the proposed ML-HDR in the ptychographic reconstructions versus the dynamic range assessed by different metrics. (a) PSNR. (b) SSIM. (c) RSM errors. (d) Truth image and the corresponding reconstruction results by different methods with 8-bit dynamic range. Herein, A, B, and C stand for the single exposure, the LRFC-HDR and the proposed ML-HDR ptychography, respectively.

imaging. Therefore, how to measure the absolutely correct raw diffraction signals with high SNR in ptychographic lensless computational imaging is the primary problem to be considered. Afterward, fast and robust calibration of the system error parameters likewise becomes a non-negligible problem. Therefore, the cross correlation method [37], auto-focusing strategy [38], and angle self-calibration method [39] are applied to the translational position error, axial distance error and reflection angle error calibrations in ptychographic microscopy, respectively.

III. NUMERICAL SIMULATIONS

To illustrate the feasibility and effectiveness of the proposed methodology, a series of numerical simulations have been performed on a lensless ptychographic system, whose detailed configurations are introduced as follows. The collimating laser is focused on the back focal plane of 256×256 pixels and the probe interacted with the “cameraman” of 1024×1024 pixels. The object is driven by x - y linear stages to scan within a 20×20 grid motion with a step size of ~ 40 pixels containing 10% random offset to avoid artifacts. A detector of 256×256 pixels with different dynamic ranges is installed on the plane 50 mm far from the “cameraman,” recording raw diffraction data and dark frames in the ptychographic simulations.

A. Simulations Versus the Dynamic Range

The first case is to investigate the performance improvement of the proposed ML-HDR method on the ptychographic imaging under different dynamic ranges of the detector.

To better simulate the practical scenario, parameters were set close to configurations of actual devices. With 10^9 phon/sec fluxes, 400 diffraction frames and 20 dark frames in different integral times were recorded by the detector with different bit depths. Meanwhile, extremely low-intensity mixed noises, consisting mainly of the PSN, DCSN, and RN, were coupled into the diffraction frames. The mixed noises (except for the PSN) of the same level were also added to the dark frames. Based on the established noise model of the detector, the PSN, DCSN, and RN, together with the diffraction signals, continued to accumulate as the integral time increased. When the accumulation exceeded the full well capacity of 2.5×10^6 electronics, the pixel was overexposed and the detector’s digital readout reached the maximum without increasing under the current bit depth.

Three different methods, including the single-exposure, the LRFC-HDR, and the proposed ML-HDR ptychography, are applied to deal with the raw data comparatively. After 250 iterations of momentum-accelerated ptychographical iterative engine (mPIE [34]), the convergence results are shown in Fig. 2. Since the root mean square (RMS) error [35], similar to the “R-factor” [36], has explicit limitations to assess the ptychographic reconstruction image quality, the structural similarity (SSIM) and peak SNR (PSNR) are also introduced as the assessment metrics for the ptychographic reconstruction image quality [40]. It can be seen from Fig. 2(a)–(c) that the image qualities are very poor under the low bit depths, and they continue to improve with the increase of the dynamic range, for all the ptychographic reconstructions based on three different methods (i.e., the single exposure, LRFC-HDR, and ML-HDR). Up to a certain bit depth, the image qualities

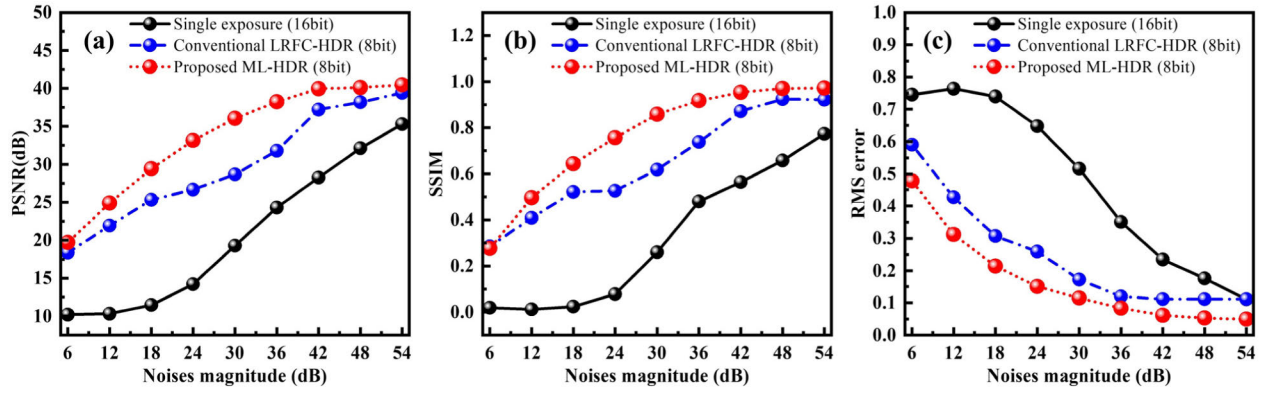


Fig. 3. Comparison of the single exposure, the LRFC-HDR and the proposed ML-HDR in the ptychographic reconstructions versus the noise level assessed by different metrics. (a) PSNR. (b) SSIM. (c) RMS errors.

remain unchanged with the dynamic range increasing. This phenomenon can be explained as the SNR of the diffraction field is very poor under low bit depth, and the high-frequency diffraction information is submerged in the noises, which reduces the effective NA of the ptychographic imaging system. With the dynamic range increasing, the high-frequency diffraction information continues to be exposed from the noises. When the effective NA expands to the entire photosensitive chip, the image qualities of reconstruction do not increase.

Further analysis of the simulation results in Fig. 2 shows that compared with the single-exposure method, the multi-exposure HDR techniques have significantly improved the image reconstruction quality of the ptychography under the same dynamic range. Moreover, compared with the conventional LRFC-HDR, the proposed ML-HDR method has further improved the image reconstruction quality. These conclusions can be further confirmed from Fig. 2(d), which shows the truth image compared with the reconstructed results via the three methods under a specific dynamic range of 8 bits corresponding to results pointed out by the gray solid lines in Fig. 2(a)–(c) marked with A, B, and C, respectively. The reason is that in the ML-HDR, the joint probability is obtained and the absolute irradiance is fused based on the maximum likelihood estimation to reduce noise weights, which is not taken into account for the conventional LRFC-HDR. On the other hand, the above conclusions can be discussed in another way that the HDR methods can effectively reduce the demand on the dynamic range of the detector. As indicated by the dashed gray lines in Fig. 2(a)–(c), a detector of 8-bit dynamic range, even lower, with the help of the proposed ML-HDR technique can achieve performances of the same level by a detector with dynamic range over 16 bits without HDR.

B. Simulations Versus the Noise Level

In the second case, simulations were performed to investigate the influences of the noise level on the performances of the ptychography using the single-exposure method, the conventional LRFC-HDR, and the proposed ML-HDR as shown in Fig. 3. It can be seen that HDR techniques (with an 8-bit dynamic range) notably improve the ptychographic reconstructions accessed by all three metrics as respectively

shown in Fig. 3(a)–(c) compared with the single-exposure method (with a 16-bit dynamic range) for all the concerned noise levels. Moreover, the proposed ML-HDR demonstrates further improvements than the LRFC-HDR under the same dynamic range of 8 bits. These results indicate that the proposed ML-HDR method has a better applicable range for the ptychography under different noise levels than the single-exposure methods and conventional LRFC-HDR.

It should be also noted that in both cases of high-level and low-level noises, improvements in the reconstruction quality are not so notable as those of the moderate-level noise, especially when using the PSNR and SSIM metrics as shown in Fig. 3(a) and (b). The phenomenon for low-level noises can be attributed to the fact that the effective NA cannot continue to increase as indicated by results in Fig. 2. While for high-level noises, the crosstalk increases, where the noises difficult to be separated from the absolute irradiance throughout, reducing the effect of the HDR method to lose its effect.

IV. EXPERIMENTS

To further validate the correctness and effectiveness of the proposed ML-HDR method, the prototypes of the ptychographic microscope in transmission and reflection geometries [41], [42] were set up to conduct experimental verifications as illustrated in Fig. 4. The collimated coherent beam of 632.8 nm was radiated by the He–Ne laser (Newport, N-STP-912) and passed through different filters, a pinhole 1 (Thorlabs, ID25Z), a beam expander (Newport, GBE05B) and a pinhole 2 (Newport, M-ID-1.0) to a plano-convex lens (Newport, KPX049AR.14) in Fig. 4(a). Then, the parallel beam was focused on the back focal plane by the lens and the focusing probe interacted with the specimen in transmission or reflection geometries, generating coherent diffraction fields. Meanwhile, an x - y linear stage (Physik Instrumente, S-C01K016) moved with the specimen in a 21×21 mesh trajectory. A back-side illuminated scientific research CMOS detector (QHYCCD, QHY600PH-SBFL) with $3.76 \mu\text{m}$ pixel size was fixed on the specimen downstream position to collect 441 diffraction frames and 20 dark frames with different exposure times. The ptychography setups in the transmission and reflection geometries are shown in the Fig. 4(b) and (c), respectively.

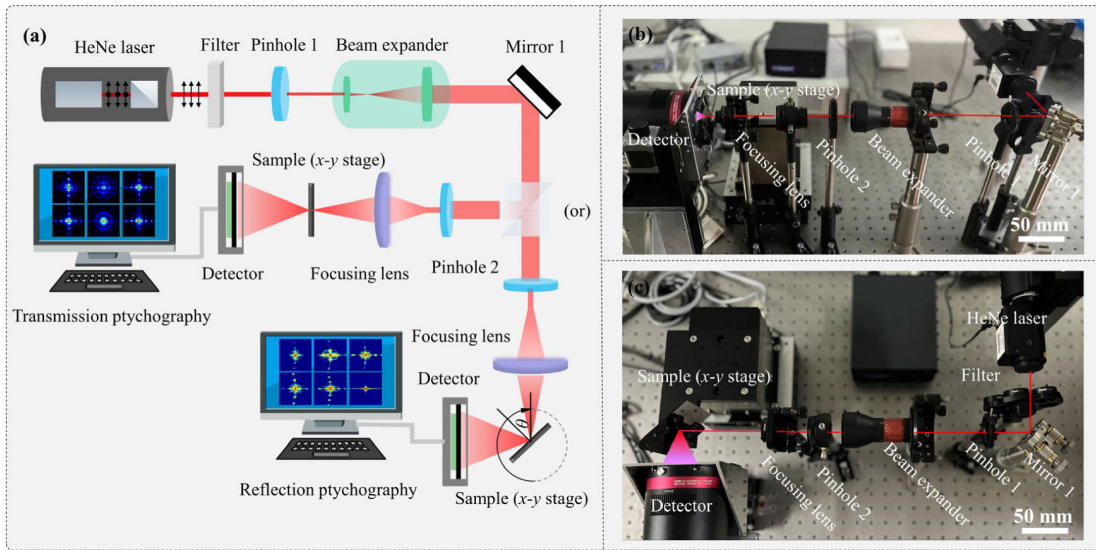


Fig. 4. Ptychographic experimental setup both in transmission and reflection geometries. (a) Schematic of the ptychography in both geometries. (b) Prototype of the ptychographic microscope in transmission geometry. (c) Prototype of the ptychographic microscope in reflection geometry.

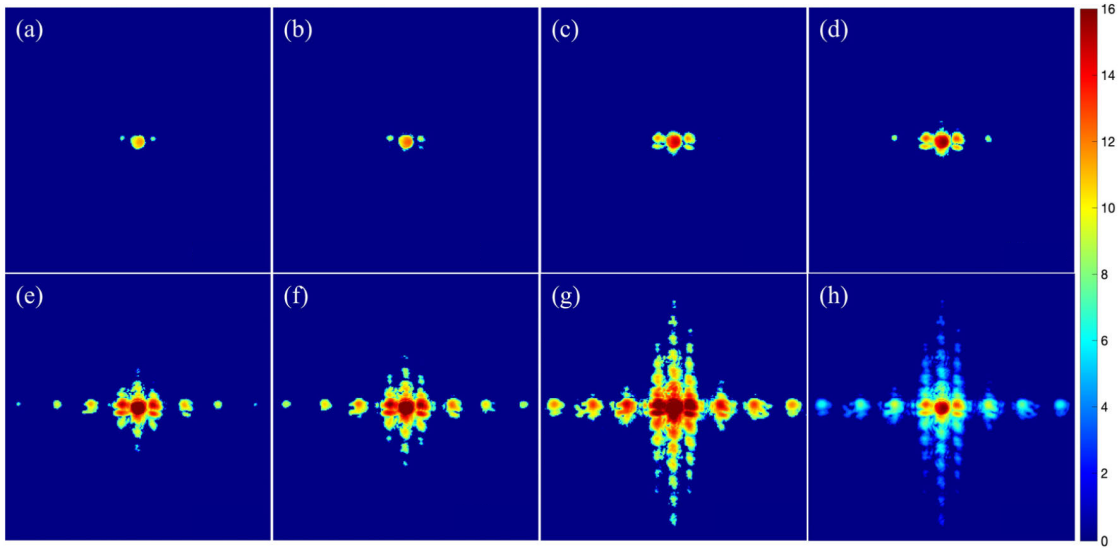


Fig. 5. Measured diffraction frames versus different exposure times. (a) 0.5 ms. (b) 1 ms. (c) 5 ms. (d) 10 ms. (e) 50 ms. (f) 100 ms. (g) 500 ms. (h) Final fused ML-HDR absolute irradiance.

A. Transmission ML-HDR Ptychographic Microscope

In the ptychographic microscope of the transmission geometry in Fig. 4(b), the linear stage moved mesh scanning with a positive resolution target USAF-1951 (Edmund optics, #58-198), and the grid gap was about $10\ \mu\text{m}$ with 10% random offset to ensure more than 80% overlap ratio. To balance the computational efficiency and the NA of the imaging system, the 4096×4096 pixels central area was cropped and binned 16 in different bit depths (8 or 16 bits), i.e., $M \times N = 256 \times 256$ pixels with $60.16\ \mu\text{m}$ pixel sampling. In fact, the various binning of detector pixels is feasible because the redundant data of ptychographic overlapping measurement can relax the sampling requirements of detector sensor, i.e., the reciprocal-space up-sampling results from the real-space oversampling in ptychographic imaging [43], which also make the ptychographic detector sampling dimensions can easily

break the sampling criterion of CDI or conventional imaging systems. After calibrating the system errors [37], [38], the ptychographic diffraction frames were collected by the CMOS detector with different exposure times and are shown in Fig. 5(a)–(g) under 8-bit dynamic range. Meantime, the dark frames were also collected with same exposure times. Based on the direct measurement parameters of the diffraction frames, dark frames and exposure times, the fusional absolute irradiance by the ML-HDR method is shown in Fig. 5(h). From Fig. 5(a)–(g), it can be clearly seen that as the exposure time increases, the diffraction information in the high Q regions gradually exposes, in the meanwhile, the low-frequency diffraction information is severely overexposed. The collected diffraction fields can only reflect its truth in a limited range. When the ML-HDR image fusion technology is applied to ptychography imaging, Fig. 5(h)

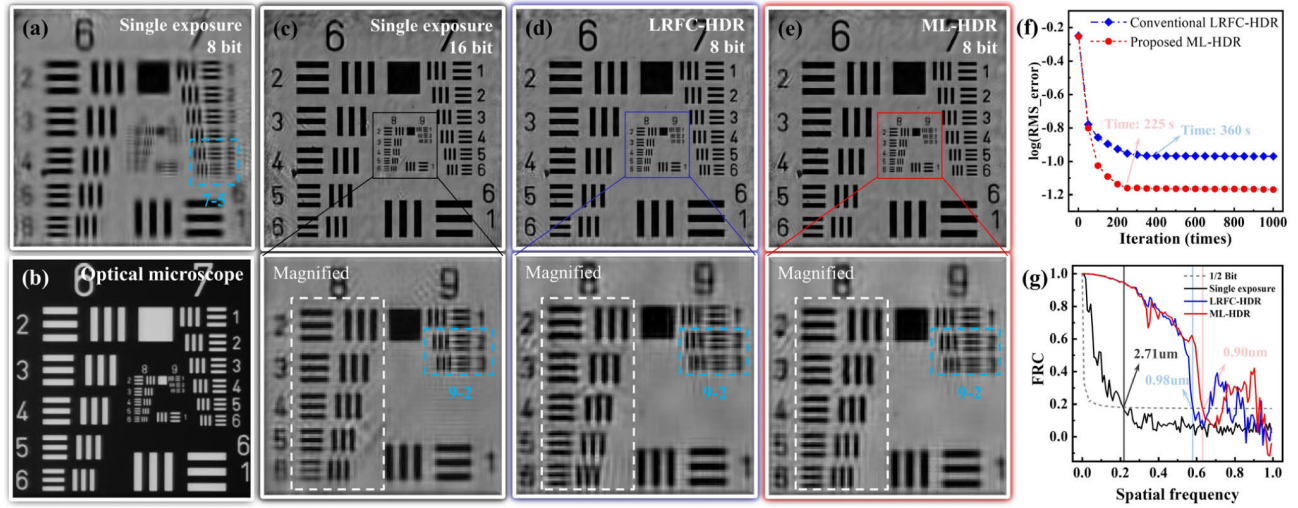


Fig. 6. Experimental results of the transmission ptychographic microscope. (a) Single exposure with 8-bit dynamic range; (b) Optical microscope; (c) Single exposure with 16-bit dynamic range; (d) Conventional LRFC-HDR with 8-bit dynamic range; (e) Proposed ML-HDR with 8-bit dynamic range; (f) Convergent rms error logarithm between the conventional LRFC-HDR and proposed ML-HDR; (g) FRC among the single exposure, conventional LRFC-HDR and proposed ML-HDR.

shows more details in the ptychography compared with single exposures.

After 1000 mPIE iterations (MATLAB, R2021a; CPU, Intel i7-12700F; GPU, NVIDIA RTX 3060 Ti), the reconstruction results are shown in Fig. 6(e). Compared with the results of 10 ms single-exposure ptychography in Fig. 6(a), the proposed method can distinguish the group 9/element 2 (Resolution: $0.87 \mu\text{m}$), while single-exposure ptychography could only distinguish group 7/element 5 (Resolution: $2.461 \mu\text{m}$). To eliminate the uncertainty in visual observation, the spatial frequency of the Fourier ring correlation (FRC) [44], [45] was introduced to quantitatively evaluate ptychographic imaging resolution as shown in Fig. 6(g). Both qualitative observations and quantitative FRC results indicated that the proposed ML-HDR delivers approximately three-fold resolution enhancement. When reconstruction results in Fig. 6(d) and (e) are focused on carefully, it can be seen that image resolutions between the conventional LRFC-HDR and the proposed ML-HDR have a negligible difference. But for the feature sharpness and background uniformity, the proposed ML-HDR ptychography has a more obvious improvement than other corresponding areas of the conventional LRFC-HDR relative to the truth image in Fig. 6(b) measured by an optical microscope (Sunny Optical, BH200M with TopuTek detector, UCMOS03100KPA). The reason for this phenomenon can be attributed to the fact that the conventional LRFC-HDR only extracts high-order diffraction information and ignores the impact of mixed noises. In contrast, the proposed ML-HDR method allocates weights reasonably, which effectively suppresses the noise crosstalk of low-order diffraction information while extracting high-order diffraction information, which will be more intuitively demonstrated in the FRC results of Fig. 6(g). Further, we introduced a widely accepted quantitation rms error logarithm [35], [36] to assess the no-reference ptychographic imaging quality. The rms error assessment also clearly shows that the proposed ML-HDR ptychography has

faster convergence speed, reduce the 1.6-folds time overhead, than the conventional LRFC-HDR ptychography in Fig. 6(f).

Other comparative experiments with different dynamic ranges were conducted and the results are shown in Fig. 6(c)–(e), and it clearly indicates that the proposed ML-HDR technique mitigates 8-bit dynamic range for the detector, namely, a detector of 8-bit dynamic range with the help of the proposed ML-HDR technique can achieve performances of the same level by a detector with dynamic range over 16 bits without HDR. But for the conventional LRFC-HDR ptychography in the Fig. 6(d), the reconstruction results of imaging sharpness, contrast, and resolution, especially in the background uniformity and the number label in magnified images of white dashed area, are worse than the single exposure ptychography with 16-bit dynamic range and proposed ML-HDR ptychography with 8-bit dynamic range, which demonstrates the superiority of the proposed ML-HDR method in mitigating dynamic range of detectors while being immune to noise.

To further explore the potential performance of the conventional LRFC-HDR and proposed ML-HDR ptychography in a higher 16-bit dynamic range of detector, comparative experiments were conducted and the reconstruction results are shown in Fig. 7(b) and (c) after 1000 mPIE iterations, and Fig. 7(a) shows the observed result by an optical microscope. It is obvious that there is a significant improvement in reconstruction resolution (group 9/element 3: $0.775 \mu\text{m}$) than the experiment with detector of 8-bit dynamic range in Fig. 6(d) and (e). In the meanwhile, the line trace approach [46], [47] and the FRC approach are also applied to assess the reconstruction image resolution in no-reference ptychography, and shown in Fig. 7(d) and (e). The line trace results clearly reveal that the element features of the proposed ML-HDR ptychography in group 9 have higher imaging sharpness and contrast than the conventional LRFC-HDR ptychography, and the FRC results quantitatively reveals

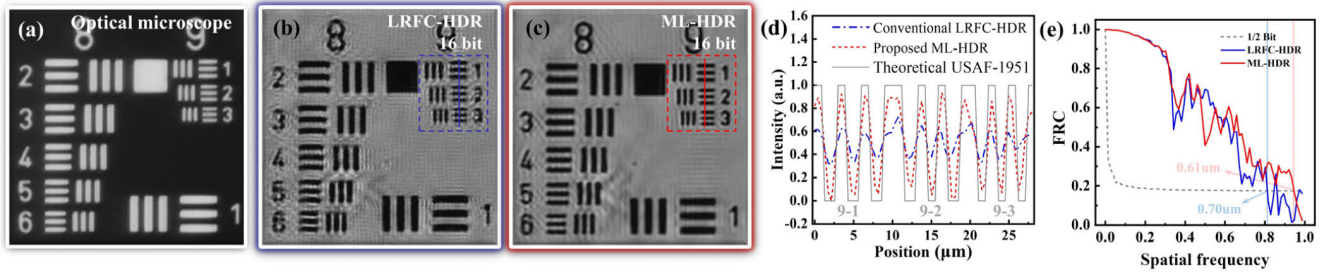


Fig. 7. Experimental results with 16-bit dynamic range. (a) Optical microscope. (b) Conventional LRFC-HDR; (c) Proposed ML-HDR; (d) Line trace between the conventional LRFC-HDR and proposed ML-HDR relative to theoretical USAF-1951. (e) FRC between the conventional LRFC-HDR and proposed ML-HDR.

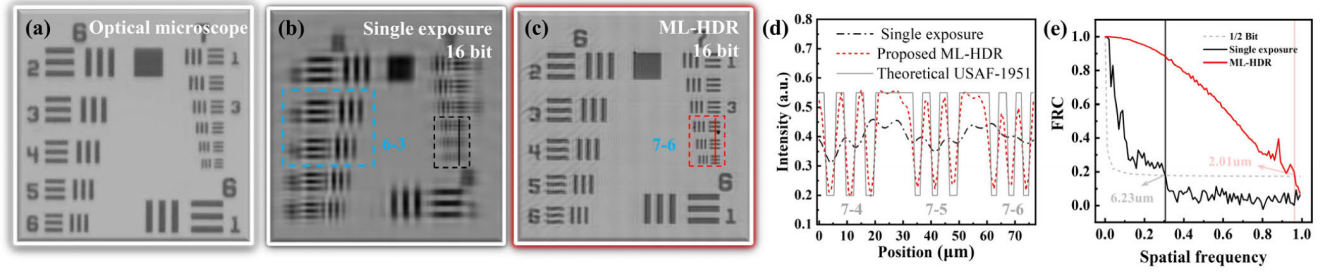


Fig. 8. Experimental results of the reflection ptychographic microscope. (a) Optical microscope. (b) Conventional LRFC-HDR. (c) Proposed ML-HDR. (d) Line trace between the conventional LRFC-HDR and proposed ML-HDR relative to theoretical USAF-1951. (e) FRC between the single exposure and proposed ML-HDR.

enhancement magnitude, approximately 1.15 folds, of the proposed ML-HDR than conventional LRFC-HDR and calibrate the limiting resolution of the ptychographic microscope, beyond the range of the norm USAF-1951 resolution target.

B. Reflection ML-HDR Ptychographic Microscope

In the ptychographic microscope of the reflection geometry in Fig. 4(c), the linear stage moved mesh scanning pattern, keeping 45° reflection geometry by angle error calibration [39], with a positive reflection target of USAF-1951 (Thorlabs, R3L3S1PR), which is a weakly scattering specimen with low contrast between the AR-coated chrome pattern and uncoated chrome background and has a harder reconstruction for lower SNRs in directly measured diffracted frames compared with transmission USAF-1951 (Edmund optics, #58-198). Due to the geometry limitations of the reflection setup, the distance of the detector from the specimen increases from 13.9 mm in transmission setup to 48.5 mm. The larger photosensitive chip area was cropped, i.e., 6144×6144 pixels, and the larger binning 24 was operated for collected raw data to balance the efficiency and resolution. After 1000 mPIE algorithm iterations, the results indicate that the ML-HDR ptychography very clearly reconstructs group 7/element 6 (Resolution: $2.192 \mu\text{m}$) in Fig. 8(c). Under the same conditions, the single exposure ptychography can only distinguish group 6/element 3 (Resolution: $6.201 \mu\text{m}$) in Fig. 8(b). Due to the influence of the sample contrast and the sampling dimension of the imaging sensor in the optical microscopy, i.e., pixel size, the imaging result in Fig. 8(a) has a few degraded compared to the ML-HDR ptychography. Further, compared with the single exposure ptychography, the ML-HDR ptychography significantly improves the imaging resolution of the weakly scattering reflection target with the 16-bit dynamic range

detector. In the line trace assessment of the Fig. 8(d) and the FRC assessment of the Fig. 8(f), the results also more intuitively reveals that the ML-HDR approach improvements, more than threefold resolution enhancement, relative to the single exposure ptychography.

Regardless of the transmission or reflection geometries, all the experimental results indicate that the ML-HDR approach is an effective resolution enhancement technique when the dynamic range of the detector in ptychographic microscope is insufficient. Compared with the single exposure ptychography, the ML-HDR approach mitigates 8 bits for the detector's dynamic range and enhances the resolution of ptychographic imaging nearly threefold. In the meantime, due to the outstanding advantage in taking mixed noise impacts into account, the proposed ML-HDR approach has a superior imaging uniformity and contrast in ptychographic reconstructions relative to the conventional LRFC-HDR ptychography.

V. CONCLUSION

In the article, a simple-to-implement and noise-robust ML-HDR image fusion approach is proposed to broaden the effective NA of ptychographic imaging. First, we constructed a joint distribution containing measured diffraction frames, dark frames, and the absolute radiances to be retrieved. Then, based on maximum-likelihood estimation of the joint probability, the weighting function was optimized to fuse the HDR absolute irradiance from a sequence of direct measurement parameters. Finally, simulations and experiments were conducted to validate the effectiveness and correctness of the proposed ML-HDR approach for ptychographic microscope in both transmission and reflection geometries. Results indicate that the ML-HDR approach significantly relaxes the 8-bit dynamic range of detectors and enhances the resolution of

ptychographic imaging by nearly threefold compared with the conventional single exposure ptychography. More importantly, the proposed ML-HDR approach does not require any additional parameter calibration for detectors and takes temporal mixed noise into account in the image fusion, which makes it easier to implement and more immune to noise than the existing LRFC-HDR approach. These remarkable advances greatly broaden the potential application fields, including infrared imaging, fluorescence imaging and photoacoustic imaging, and so on.

Here, only the temporal mixing noises of the detector are briefly analyzed and the spatial mixed noises, including the non-uniformity of photo-response and the non-uniformity of dark current [30], [48], [49], are roughly eliminated by the background subtraction between diffraction frames and dark frames, which ignores and simplifies the dynamic changes of the spatial noises with integral time increase. Therefore, a more complete and accurate spatiotemporal mixed noise model will be focused on to further improve the absolute irradiance accuracy in the HDR image fusion in the future.

ACKNOWLEDGMENT

The authors would like to thank the technical support from the Experiment Center for Advanced Manufacturing and Technology in School of Mechanical Science & Engineering of HUST.

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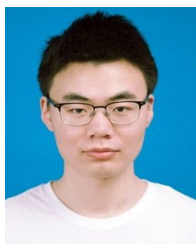
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